Exploration of FELLOW trait dataset

The objective of this document is to:

- visually explore the trait database
- understand trait coverage / data gaps
- check for possible inconsistencies

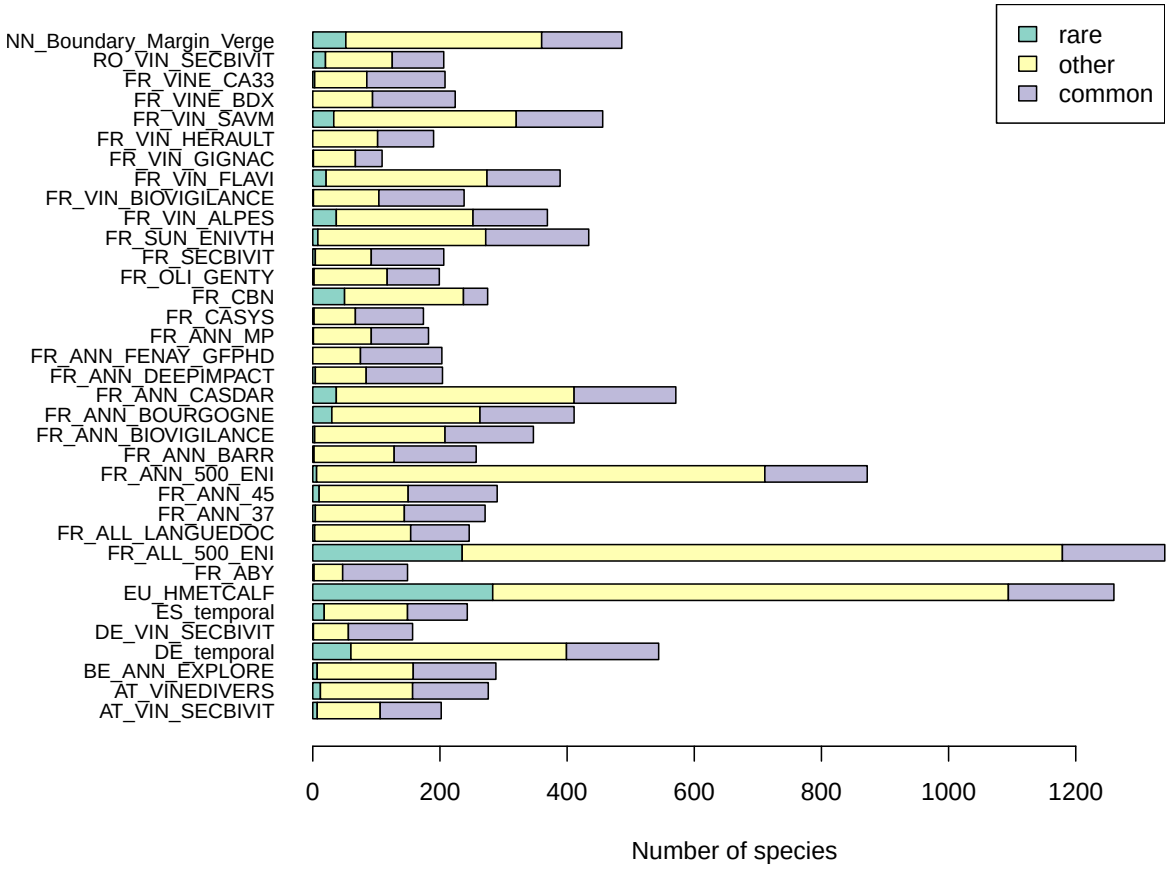
Description of the species list

We compiled the species lists from 35 datasets. After cleaning and harmonization, there were 2476 unique taxa.

```
< table of extent 0 >
```

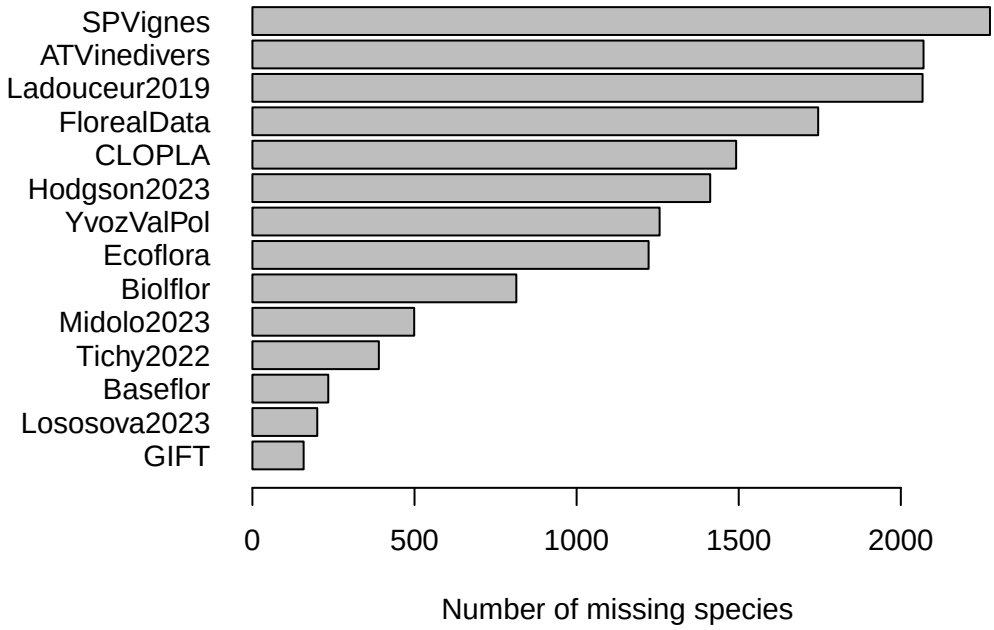
Let's define:
rare: a taxa that occurred only in a singe database
common: a taxa that is listed in 50% of the databases (18 out of 35)

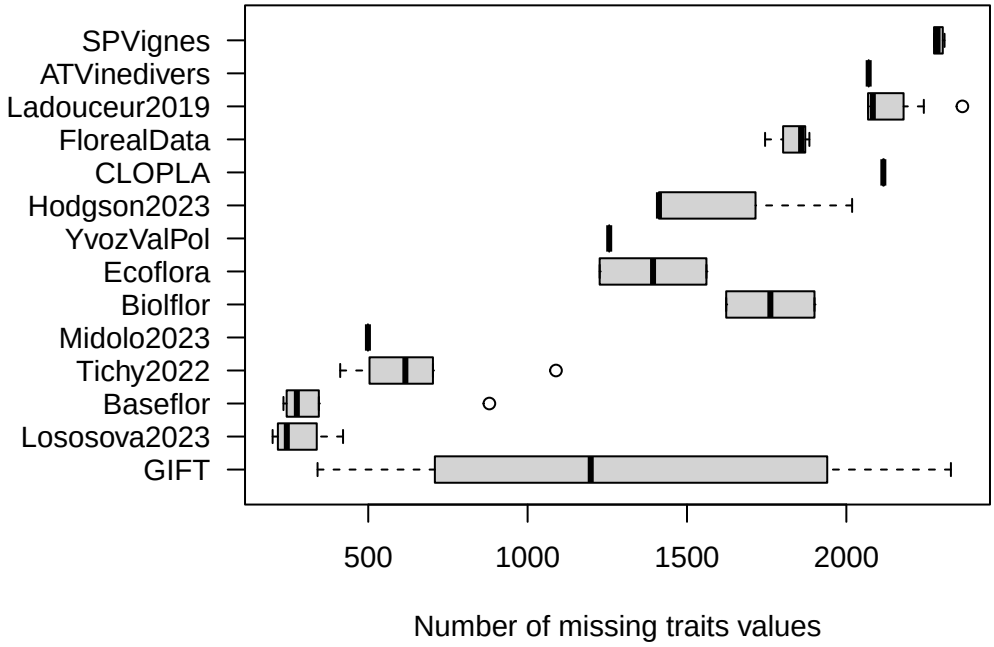
```
sp_class
  rare  other common
   959   1331    167
```

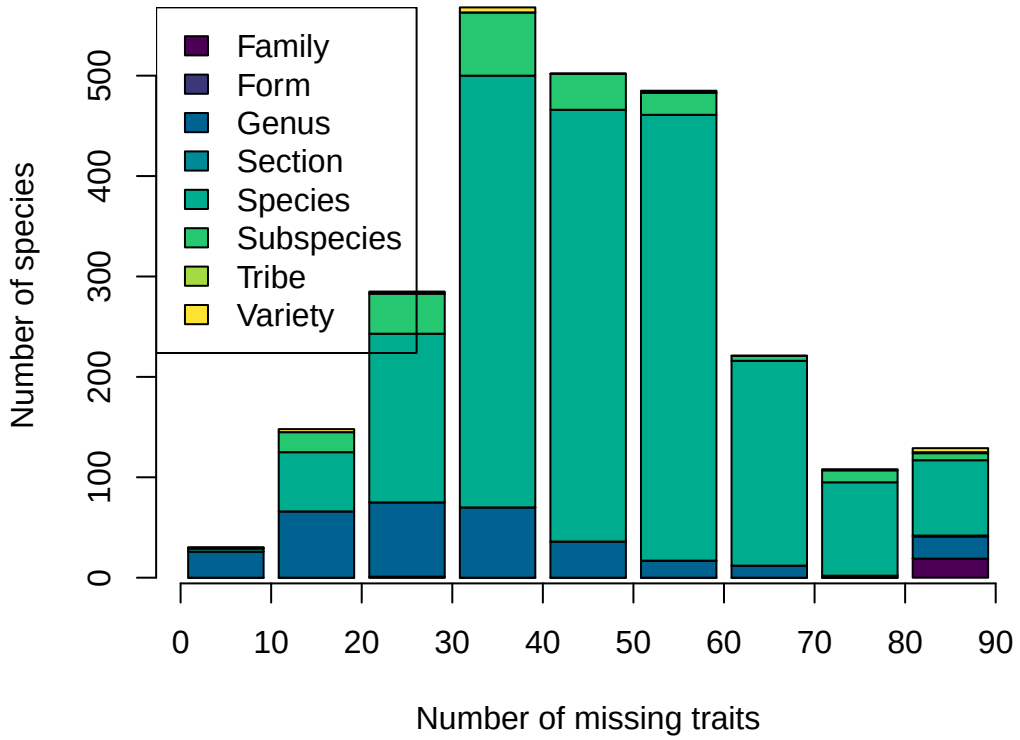


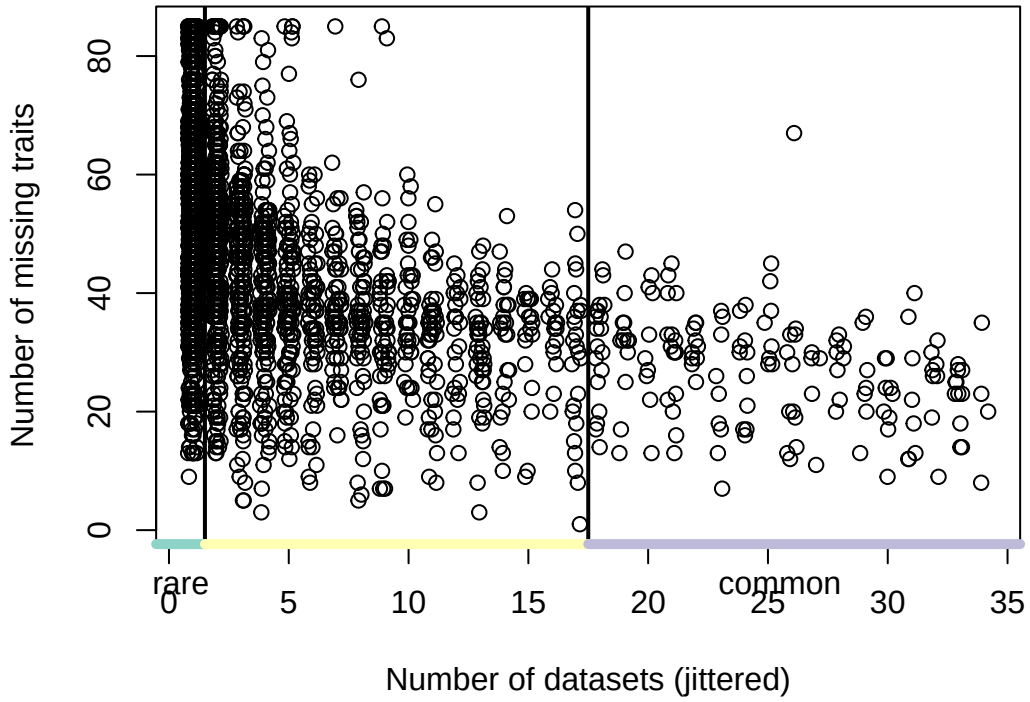
Description of trait databases

So far, we compiled 85 traits for 2476 taxa gathered from 14 trait databases. But there are many missing values.






Family
Form
Genus
Section
Species
Subspecies
Tribe
Variety
Number of species
0
100
200
300
400
500
0
10
20
30
40
50
60
70
80
90
Number of missing traits


Number of missing traits
0
20
40
60
80
rare
common
0
5
10
15
20
25
30
35
Number of datasets (jittered)

Taxa with no or limited trait information (N=82).

```
 [1] "Abies"                                "Acacia"
 [3] "Agropyron"                            "Amaranthaceae"
 [5] "Anagallis arvensis caerulea"          "Anagyris"
 [7] "Apiaceae"                             "Asparagaceae"
 [9] "Boraginaceae"                         "Brachythecium rutabulum"
[11] "Brassicaceae"                         "Bromeliaceae"
[13] "Camelina neglecta"                    "Cardueae"
[15] "Caryophyllaceae"                      "Chaenomeles x superba"
[17] "Chamomilla recutita"                  "Chenopodiaceae"
[19] "Circaea"                              "Cirsium lanceolatum"
[21] "Cochlearia"                           "Conyzigeron"
[23] "Crataegus kyrtostyla domicensis"      "Crepis rhoeadifolia"
[25] "Cupressus"                            "Cynoglossum cheirifolium"
[27] "Duchesnea indica"                     "Elymus trachycaulus"
[29] "Elytrigia intermedia trichophora"     "Epilobium x intersitum"
[31] "Ficaria verna ficariiformis"          "Galactites elegans"
[33] "Gemmabryum dichotomum"                "Geraniaceae"
[35] "Holcus hybridus"                      "Imbribryum subapiculatum"
[37] "Kandis perfoliata"                    "Lamiaceae"
[39] "Lasiospora hirsuta"                   "Lavandula"
[41] "Leontodon taraxacoides taraxacoides" "Liliaceae"
[43] "Lilium"                               "Marchantia polymorpha ruderalis"
[45] "Matricaria recutita"                  "Matricaria recutita recutita"
[47] "Matthiola"                            "Melilotus alba"
[49] "Milium"                               "Moehringia"
[51] "Muhlenbergia"                         "Myosotis palustris"
[53] "Nigella gallica"                      "Orchidaceae"
[55] "Orchis"                               "Orobanche picridis"
[57] "Paronychia"                           "Pelargonium"
[59] "Platycapnos spicatus"                 "Poaceae"
[61] "Potentilla neumanniana"               "Primulaceae"
[63] "Psoralea"                             "Rhizogemma staphylina"
[65] "Riccia"                               "Riccia sorocarpa"
[67] "Riccia warnstorfii"                   "Rosaceae"
[69] "Rubiaceae"                            "Rubus idenus"
[71] "Salix fragilis"                       "Scorzonera sect podospermum"
[73] "Senecio vernalis"                     "Thapsia"
[75] "Trigonella corniculata"               "Triticale x rimpaui"
[77] "Ulmaceae"                             "Vaccaria hispanica hispanica"
[79] "Verbesina montevidensis"              "Vicia nigra"
[81] "Vulpia myuros myuros"                 "Xanthium cavanillesii"
```

Open question:

How to deal with families taxa?

How to deal with missing trait values?

Summary of trait completness

Trait	N database	N taxa	Completness (%)
Dispersal.mode	4	2342	95
Plant.height	7	2317	94
Dispersal.distance	1	2243	91
Chorology	1	2242	91
Pollination	1	2232	90
Lifecycle	2	2221	90
Flower.color	4	2204	89
Inflorescence	2	2204	89
Seed.mass	5	2151	87
Ellenberg.Salinity	1	2064	83
Photosynthetic.pathway	1	2055	83
Disturbance.Frequency	1	1977	80
Disturbance.Frequency.herblayer	1	1977	80
Disturbance.Severity	1	1977	80
Disturbance.Severity.herblayer	1	1977	80
Grazing.Pressure	1	1977	80
Mowing.Frequency	1	1977	80
Soil.Disturbance	1	1977	80
Ellenberg.Light	1	1972	80
Ellenberg.Reaction	1	1941	78
Ellenberg.Moisture	1	1778	72
Ellenberg.Nutrients	1	1773	72
Pollination.syndrome	3	1720	69
SLA	3	1680	68
Flowering	1	1596	64
Flower.UV.reflectance	3	1465	59
Ellenberg.Temperature	1	1387	56
Leaf.area	1	1288	52
Floral.symmetry	1	1220	49
Flower.class	1	1220	49
Flower.type	1	1220	49
Nectar.quantity	1	1220	49
Pollen.quantity	1	1220	49
PV.Bees	1	1220	49
PV.Bumblebees	1	1220	49
PV.butterflies	1	1220	49
PV.Hoverflies	1	1220	49
Lifeform	2	802	32
Flower.length	2	588	24

Trait	N database	N taxa	Completness (%)
Flowering.onset	2	543	22
Vegetative.propagation	1	458	18
Clonal.index	2	431	17
Flowering.duration	1	408	16
Seeding.onset	1	408	16
Seeding.season	1	408	16
Growth.form	1	406	16
Plant.lifespan	1	406	16
Strategy.Grime	1	406	16
Clonal.presence	1	381	15
lateral.spread	1	361	15
offspring	1	360	15
offspring.wsmall	1	360	15
Seed.dormancy	1	233	9
Strategy	1	179	7
Flower.width	2	169	7
Radial.growth	1	112	5

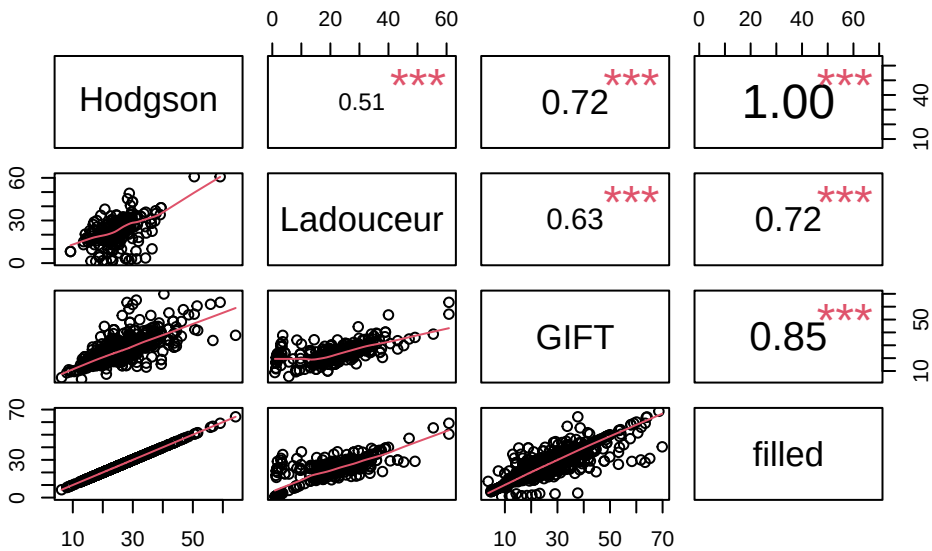
Comparison and merge

SLA

There are four sources of information for Specific leaf area (SLA) : Hodgson et al. 2023 (in mm2/mg), Ladouceur et al. 2019 (in cm2/g), GIFT (in cm2/g), and BIEN (in m2/kg = mm2/mg).

To merge and fill the SLA values (in m2/kg), we used by order of preference:

- Hodgson et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN



Number of NAs:

```
Hodgson Ladouceur      GIFT    filled
   1412      2121      1064       796
```

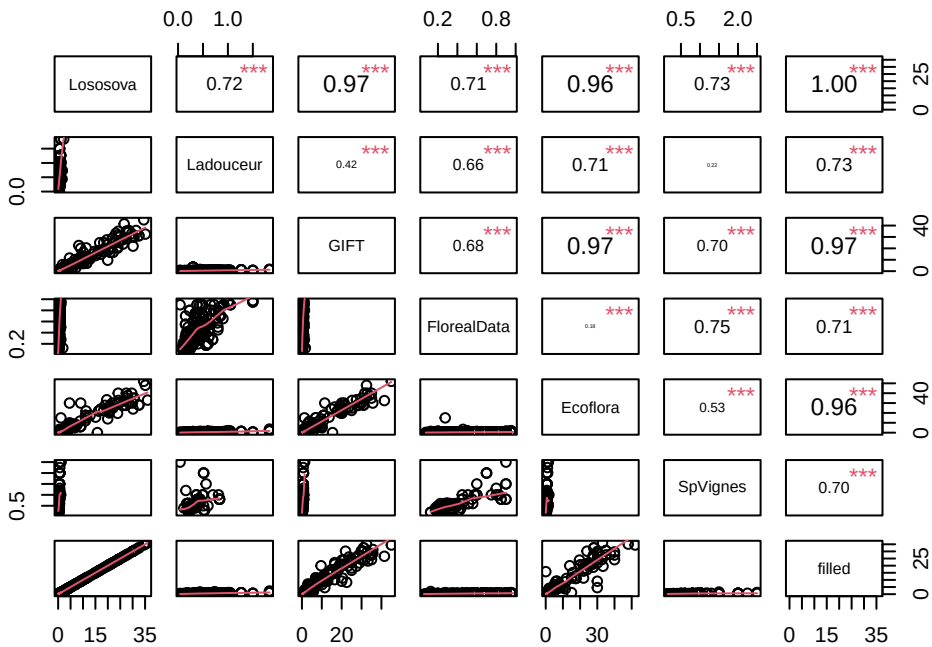
Values are highly correlated, but values from *Ladouceur et al. 2023* are only weakly correlated to other data sources (r<0.65) maybe due to different measuring protocol (?).

Plant height

There are seven sources of information for plant height. SPVignes, FlorealData, and Ladouceur_2019 are limited to small plants (<1m) (no trees) *and FlorealData>100cm must be clean.*

To merge and fill the plant height values (in m), we used by order of preference:

- Lososova et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN
- FlorealData
- Ecoflora
- SpVignes



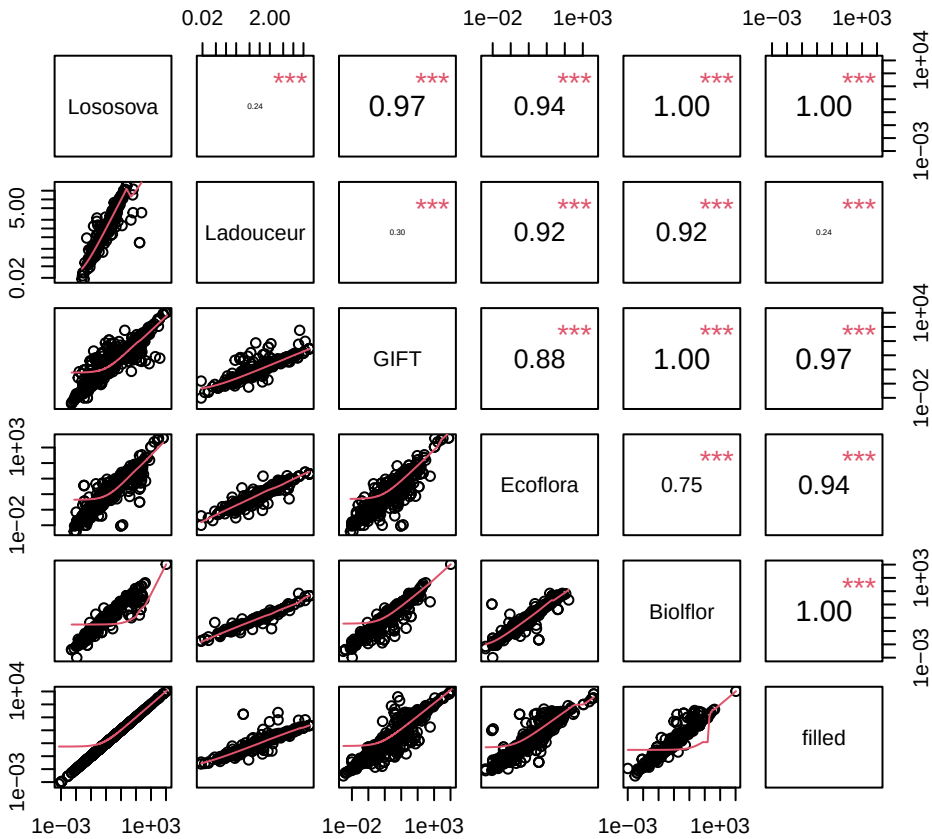
Number of NAs:

```
Lososova   Ladouceur        GIFT FlorealData    Ecoflora    SpVignes
     256        2081         709        1988        1226        2308
  filled
     194
```

Seed mass

There are six sources of information for seed mass. To merge and fill the seed mass plant height values (in mg), we used by order of preference:

- Lososova et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN
- Ecoflora
- Biolflor



Number of NAs:

```
 Lososova Ladouceur       GIFT  Ecoflora  Biolflor    filled
      421      2083        477      1561      1900       325
```

Flower colour

There are five sources of information for flower colour, but it must be cleaned before colors can be merged.

```
   Baseflor        GIFT FlorealData  YvozValPol
        345        2145        1858        1256


baseflor
             Blanc       Blanc, jaune Blanc, jaune, bleu Blanc, jaune, rose
               433                 37                  4                  4
       Blanc, rose  Blanc, vert, rose               Bleu        Bleu, blanc
                52                  2                172                 11
 Bleu, blanc, rose        Bleu, jaune  Bleu, jaune, rose         Bleu, rose
                12                  6                  2                 12
             Jaune        Jaune, rose        Jaune, vert             Marron
               542                  4                  1                 38
              Noir               Rose               Vert         Vert, bleu
                 2                341                291                 16
 Vert, jaune, rose         Vert, rose
                 1                 35
```

Dispersal mode

There are three sources of information for dispersal mode, but it must be harmonized before they can be merged.

```
Lososova2023       GIFT_1       GIFT_2     Baseflor
         200         1208         1940          275


Lososova2023
                  Anemochory                  Dyszoochory
                         344                           25
                Endozoochory                  Epizoochory
                         120                           99
                  Hydrochory Local non-specific dispersal
                           8                         1431
                Myrmecochory
                         213


GIFT
   Anemochorous Anthropochorous     Autochorous  Endozoochorous   Epizoochorous
            172              77              88              28              20
   Hydrochorous  Myrmecochorous   Unspecialized      Zoochorous
             25               3              13              82


Baseflor
               Anémochore                 Autochore                 Barochore
                      554                        33                       635
             Dyszoochore              Endozoochore Endozoochore, dyszoochore
                       10                       136                        10
             Épizoochore                Hydrochore              Myrmécochore
                      631                        54                        70
```

Inflorescence

There are two sources of information for inflorescence. GIFT database considers 11 categories, while Baseflor considers 39 categories. So the categories must be harmonized before merging.

```
    GIFT Baseflor
    2150      277


GIFT
      Catkin         Cyme      Panicle Pseudoraceme       Raceme        Spike
           5           24          152            3           45           51
       Umbel
          41


Baseflor
               Capitule de capitules                      Capitule simple
                                   9                                   37
                               Cône                             Corymbe
                                   8                                   40
                Corymbe de capitules                            Cyathe
                                  54                                   35
                          Cyme bipare                    Cyme biscorpioïde
                                 116                                    9
                   Cyme capituliforme                        Cyme d'épis
                                  27                                    7
                       Cyme d'ombelles                    Cyme de capitules
                                  30                                   29
                   Cyme de glomérules                       Cyme multipare
                                  11                                   39
              Cyme unipare hélicoïde              Cyme unipare scorpioïde
                                  97                                    6
                      Épi d'épillets                     Épi de capitules
                                 101                                   28
              Épi de cymes triflores                          Épi simple
                                  68                                   84
            Fleur solitaire latérale            Fleur solitaire terminale
                                  40                                   15
                         Glomérules                Glomérules spiciformes
                                 136                                   26
                 Ombelle d'ombellules                       Ombelle simple
                                  97                                   53
               Ombelle simple d'épis                   Panicule d'épillets
```

6
Panicule spiciforme
16
Racème d'épis
30
Racème de capitules
174
Racème de cymes unipares hélicoïdes
29
Racème de racèmes
40
Spadice
7
160
Racème capituliforme
93
Racème d'ombelles
4
Racème de cymes bipares
15
Racème de cymes unipares scorpioïdes
38
Racème simple
357

Lifecycle

There are two sources of information for lifecycle. The categories are quite coherent among GIFT and Hodgson2023 if `Herb.peren` and `Woody.peren` are merged into `Perennial`.

```
Hodgson2023        GIFT
       1412         341
```

```
             GIFT
Hodgson2023   Annual Biennial Perennial Variable <NA>
  Ann            467        1         8        0   69
  Bienn            1       28         1        0   25
  Herb.perenn      2        2       361        0   23
  Woody.peren      0        0        13        0    0
  <NA>           408       32       716        3  316
```

Clonal index

There are two sources of information for clonal index (Ladouceur 2019 and CLOPLA) but most of them are incomplete. The information seems coherent among the two databases. The order of preference is Ladouceur 2019 then CLOPLA.

```
Ladouceur2019        CLOPLA        filled
         2238          2119          2045
```

```
             CLOPLA
Ladouceur2019    1    2    3    4    5    6    7 <NA>
         2       0    7    0    0    0    0    0    8
         3       0    0   40    0    2    0    0   25
         4       0    0    0   49    0    0    0   28
         5       0    0    0    1   32    0    0   15
         6       0    0    0    1    0    9    0    5
         <NA>    3    5   45   50   65   17    6 2063
```